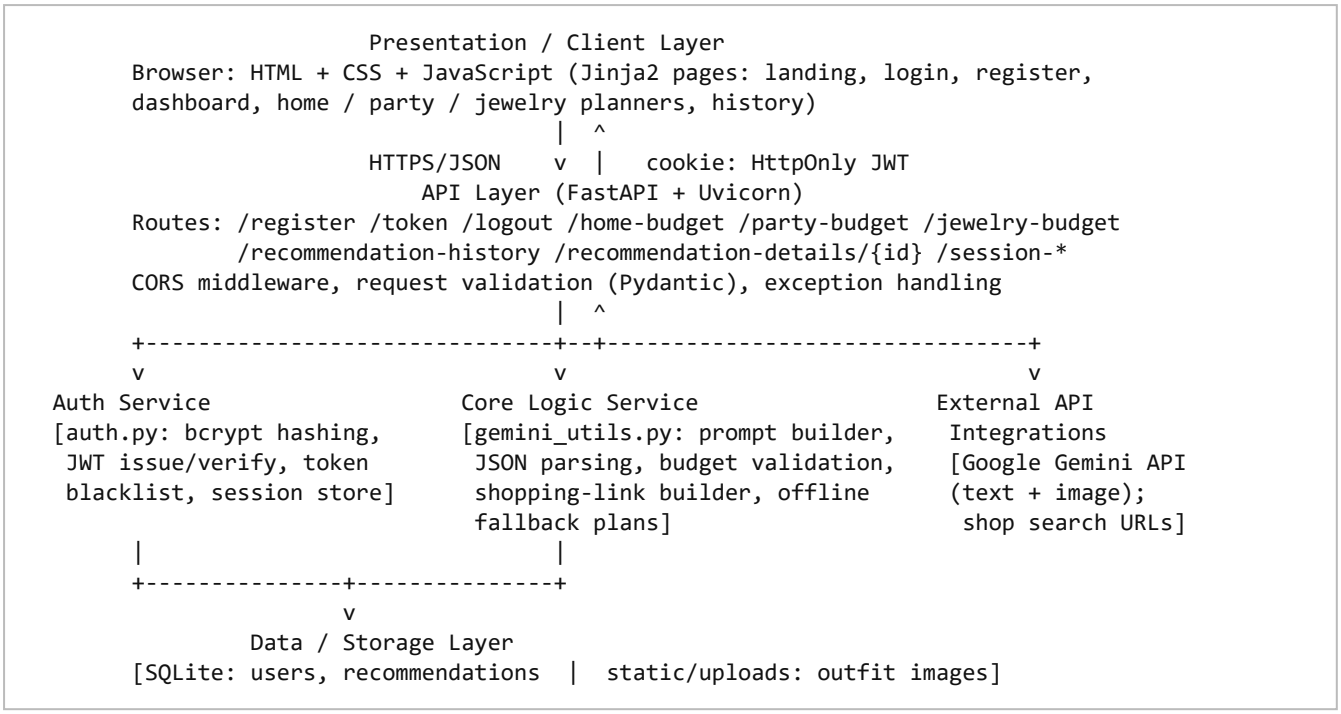
Solution Architecture

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Responsive pages and forms; renders plans as tables and cards; history modal; print view	HTML5, CSS3, JavaScript, Jinja2
API Layer (`main.py`)	Routing, validation, sessions, CORS, error handling, aliases `/generate-home`, `/generate-party`, `/generate-jewelry`	FastAPI, Uvicorn, Pydantic
Auth Service (`auth.py`)	Hashes passwords, issues and verifies JWTs, blacklists tokens on logout, tracks active sessions and cleans up idle ones	bcrypt, python-jose
Core Logic / AI Service (`gemini_utils.py`)	Builds prompts, calls Gemini with model fallback, parses JSON, recomputes totals, trims to budget, adds shopping links, supplies offline fallback	google-genai, Pillow
Models (`models.py`)	Input and session schemas with edge-case validation	Pydantic
Database (`database.py`)	Persists users and saved plans (schema with foreign key and index)	SQLite
External APIs	Gemini for recommendations; platform search URLs for shopping	Google Gemini API, URL templates